

ASL 757 Mini Project

Monal Gupta
2021TT10302

08/04/2025

To find the latitude of ITCZ for summer and winter seasons from the given rainfall data and plot it on a map.

The Intertropical Convergence Zone (ITCZ) is a band near the equator where trade winds converge, resulting in heavy rainfall. Its position shifts seasonally with solar heating, moving northward during summer and southward in winter. This report analyzes gridded rainfall data to locate the ITCZ position during DJF and JJA.

Dataset Monthly Precipitation Dataset of time period 2010 with Spatial Resolution: $0.25^\circ \times 0.25^\circ$

Tools used to calculate latitude of ITCZ: Python, Xarray, NumPy, Cartopy, Matplotlib, SciPy

Code Used: [monalgupta03/ASL757_MiniProject](https://github.com/monalgupta03/ASL757_MiniProject)

itcz_analysis.py calculates and returns the ITCZ values for Summers and Winters.

Itcz_map.py plots the two maps required

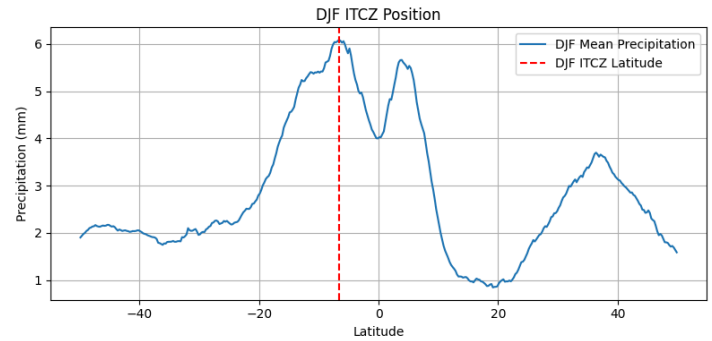
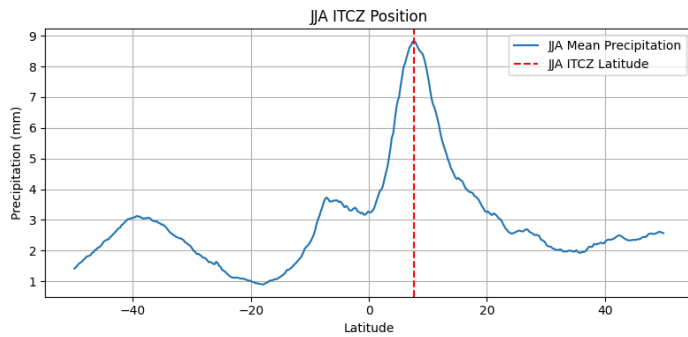
Methodology

Used xarray to extract pr variable. Separated the data into seasonal segments: DJF and JJA. Averaged rainfall values across both time and longitude to get zonal mean rainfall for each latitude. Identifying the latitude with maximum zonal rainfall in each season — considered the ITCZ latitude.

```
● PS C:\semesters\sem 8\ASL757\proj> python itcz_analysis.py
>>
DJF ITCZ Latitude: -6.625
JJA ITCZ Latitude: 7.625
```

These values align with how the Intertropical Convergence Zone (ITCZ) shifts seasonally:

- During DJF, the sun is over the Southern Hemisphere, so the ITCZ shifts south.
- During JJA, the sun is over the Northern Hemisphere, so the ITCZ moves north.



To plot the graphs,

Rainfall is averaged over time. The ITCZ is calculated as the latitude of maximum rainfall at each longitude. Applied Gaussian smoothing to remove short-scale noise. Applied latitude bounds (e.g., within 20° of equator) to ignore anomalous spikes. Converted longitudes from 0–360° to –180–180° for easier interpretation.

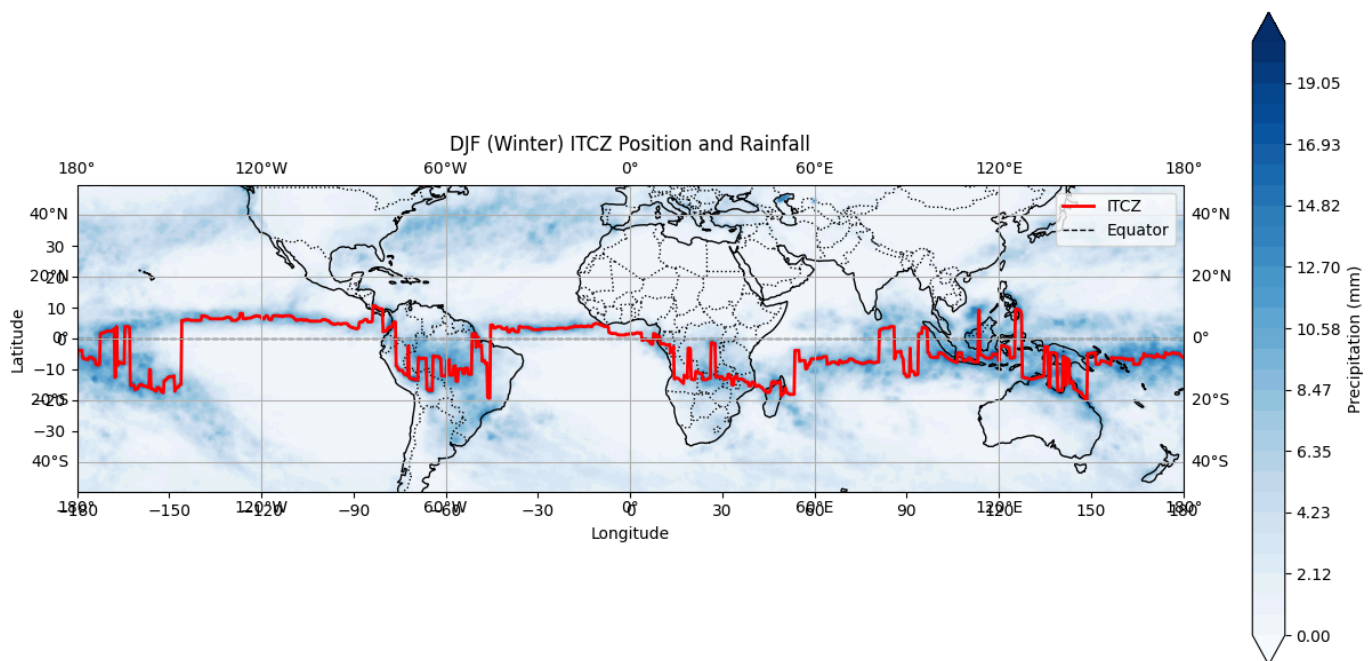
In the code,

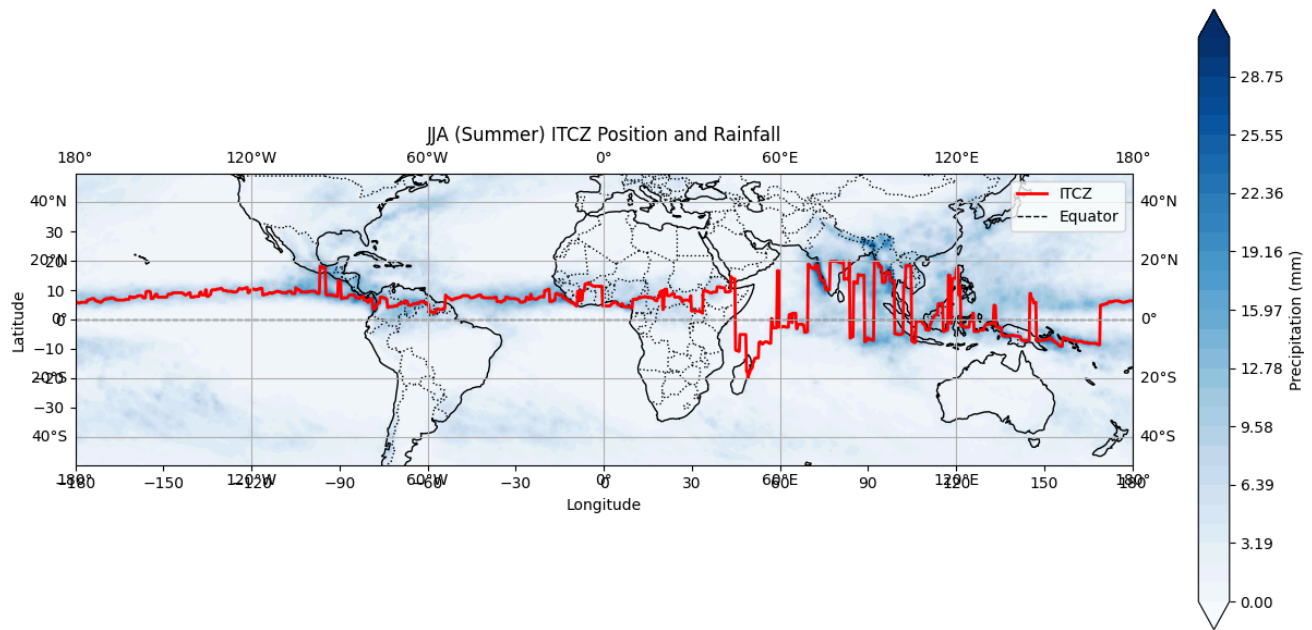
`djf_smoothed_sub` contains **smoothed rainfall values** within the $\pm 20^\circ$ latitude band.

`np.nanargmax(..., axis=0)` finds the **latitude index** where rainfall is **maximum** for each longitude.

That index is then mapped to actual latitude using: `djf_itcz=lat_subset[djf_itcz_idx]`

Using Cartopy and Matplotlib, the following plots were generated:





- Red line: ITCZ
- Blue shades: Rainfall magnitude
- Black dashed line: Equator

Conclusion

The ITCZ shows clear seasonal migration across the equator. By analyzing precipitation intensity and identifying zones of maximum rainfall, this method accurately tracks ITCZ movement. Smoothing and spike filtering ensure reliable and realistic paths.